Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 15

Chylomicrons, VLDL and LDL Metabolism

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Lecture: Chylomicrons, VLDL and LDL Metabolism

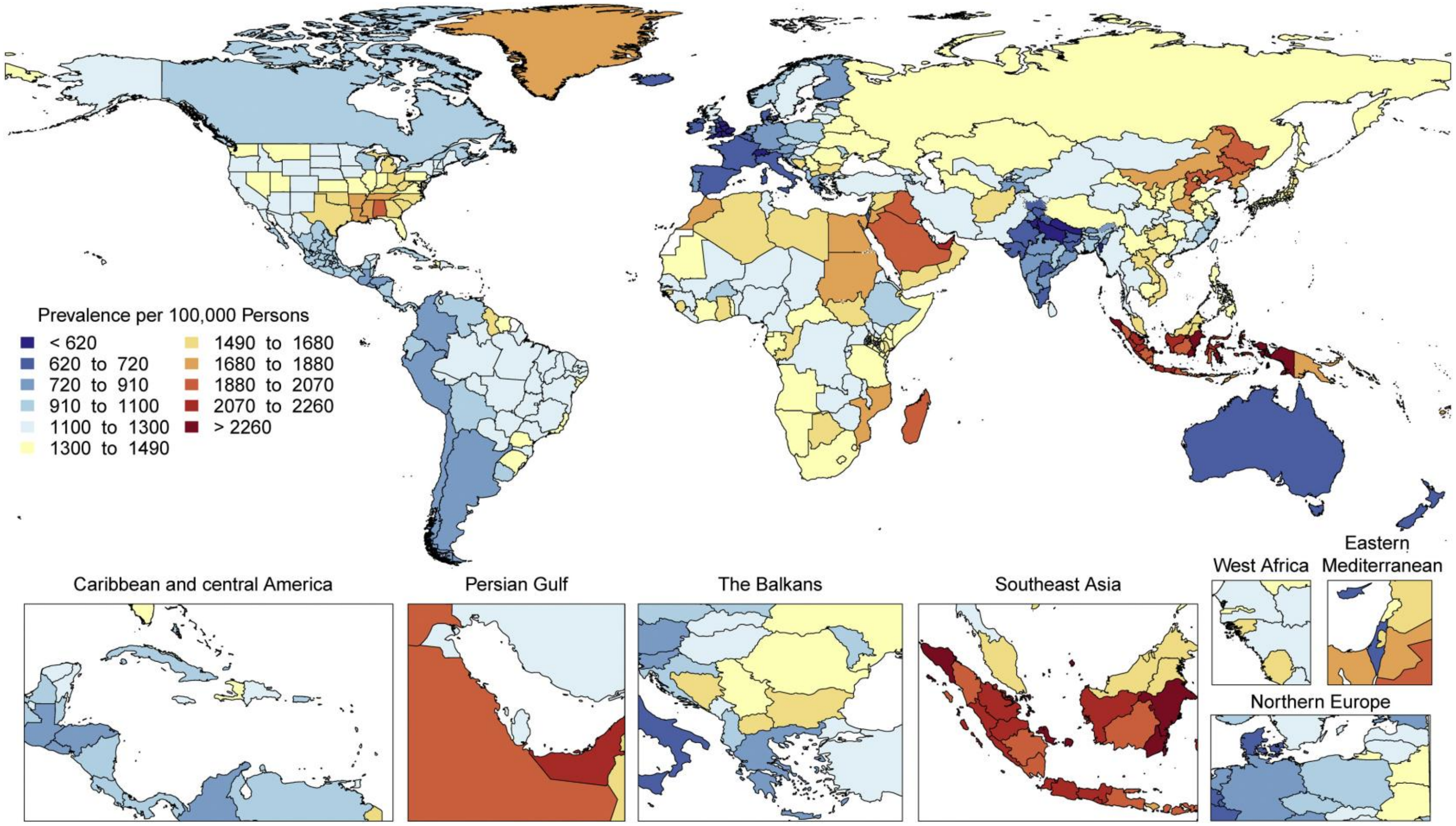
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0215	Discuss lipid transport in blood and the role of lipoproteins and albumin.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0216	Discuss chylomicron metabolism starting from formation to its uptake into the liver.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0217	Discuss the transport of dietary (exogenous) TAGs from the intestine to the peripheral tissues by chylomicrons
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0218	Describe the location and function of lipoprotein lipase in heart, skeletal muscle and adipocytes.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0219	Describe the conversion of VLDL into IDL and LDL and highlight the significance of lipoprotein lipase and hepatic lipase.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0220	Classify lipoproteins based on size and percentage of TAGs and cholesteryl esters.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0221	Outline and discuss uptake of LDL into liver and extra-hepatic tissues by receptor mediated endocytosis via the LDL receptor and remnant receptors.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0222	Describe the regulation of LDL-receptor synthesis.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0224	Explain how oxidized LDL particles are formed and how they are involved with foam cell development and atherosclerosis.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0230	Estimate LDL-cholesterol and VLDL-cholesterol using the Friedewald equation.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0233	Describe mechanism of action of statins and PCSK-9 inhibitors in hypercholesterolemia.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0234	Discuss the significance of Lp(a) and LDL-B as risk factors for coronary heart disease.

Recommended Reading

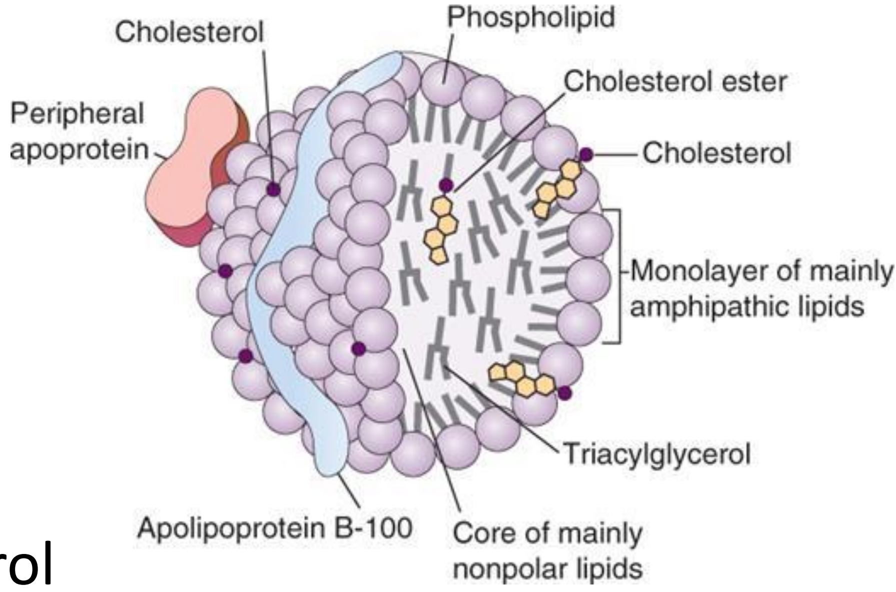
- Lippincott's Biochemistry 9th Edition
  🔗 https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402
  - Chapter 18: Cholesterol, lipoprotein and steroid metabolism; Section: Lipoproteins
- Lippincott Illustrated Reviews (**Lippincott**) 9th edition; Chapter 18; Section: Lipoproteins; Cholesterol, Lipoprotein, and Steroid Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library **Study questions: 18.2; 18.3; 18.4**
- Mark's Basic Medical Biochemistry: A clinical approach **(MBMB)** 6th edition; Sections: V, VI, VII, VIII; Cholesterol Absorption, Synthesis, Metabolism, and Fate | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library
- Rapid review Pathology: Goljan https://www.clinicalkey.com/student/content/book/3-s2.0-B9780323476683000107

Atherosclerotic Cardiovascular disease (ASCVD)

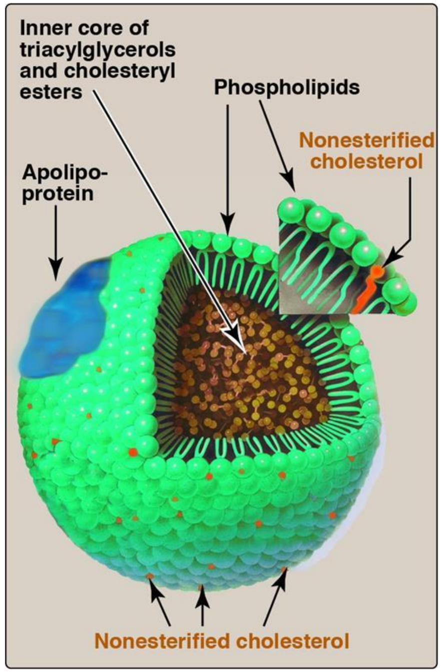
- Linked to sedentary lifestyle, obesity, dietary factors, hypertension, smoking, alcohol use, type-2 diabetes mellitus and **dyslipidemia**
- With increase in obesity, there is an increase in ASCVD
- **Dyslipidemia** closely linked to lipoprotein metabolism



Lipoproteins: General structure and function



- TAG and cholesterol ester hydrophobic (non-polar) lipids
  - Serum albumin transports free fatty acids
- Lipoproteins are spherical particles
  - Outer shell: single layer of Phospholipids, free cholesterol and apolipoproteins
  - Inner core: TAG or Cholesterol ester or both (non-polar)
- Based on composition classified as
  - Triglyceride rich/TAG-rich: Chylomicrons and VLDL
  - Cholesterol-rich: LDL and HDL
- Serum fasting lipid profile:
  - Serum Triglycerides: TAG in Chylomicrons and VLDL
  - Serum Cholesterol: Cholesterol in **LDL** and HDL and VLDL

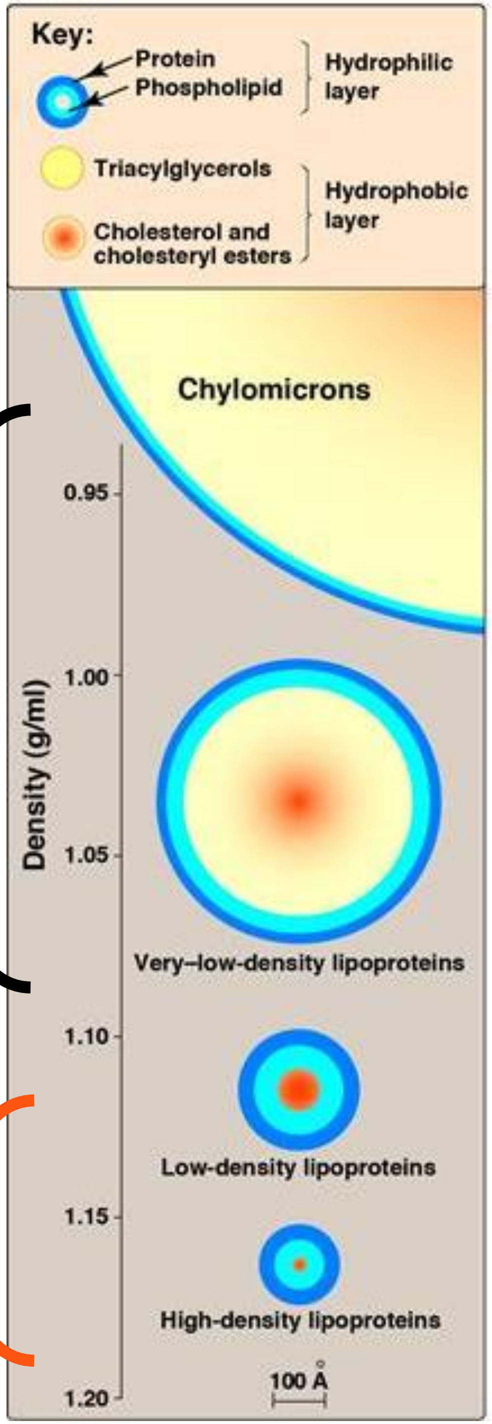


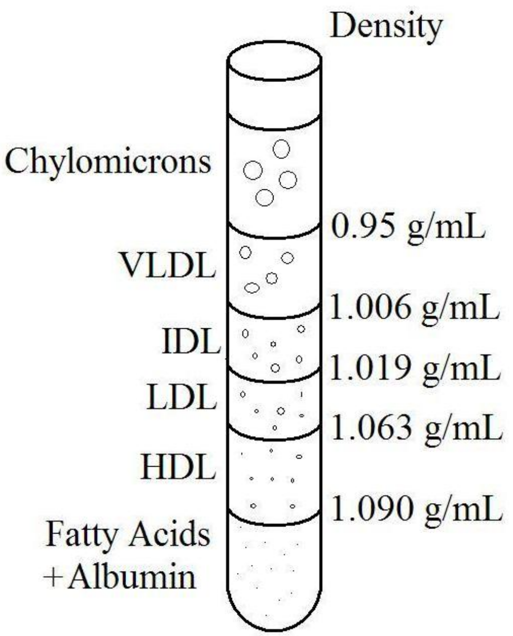
Classification of lipoproteins: Based on density (ultracentrifugation)

- **Chylomicrons (Least dense and largest)**
- **VLDL (Very low-density lipoprotein)**
- IDL (Intermediate density lipoprotein)
- **LDL (Low density lipoprotein) and**
- **HDL (High density lipoprotein and smallest)**

TAG-rich

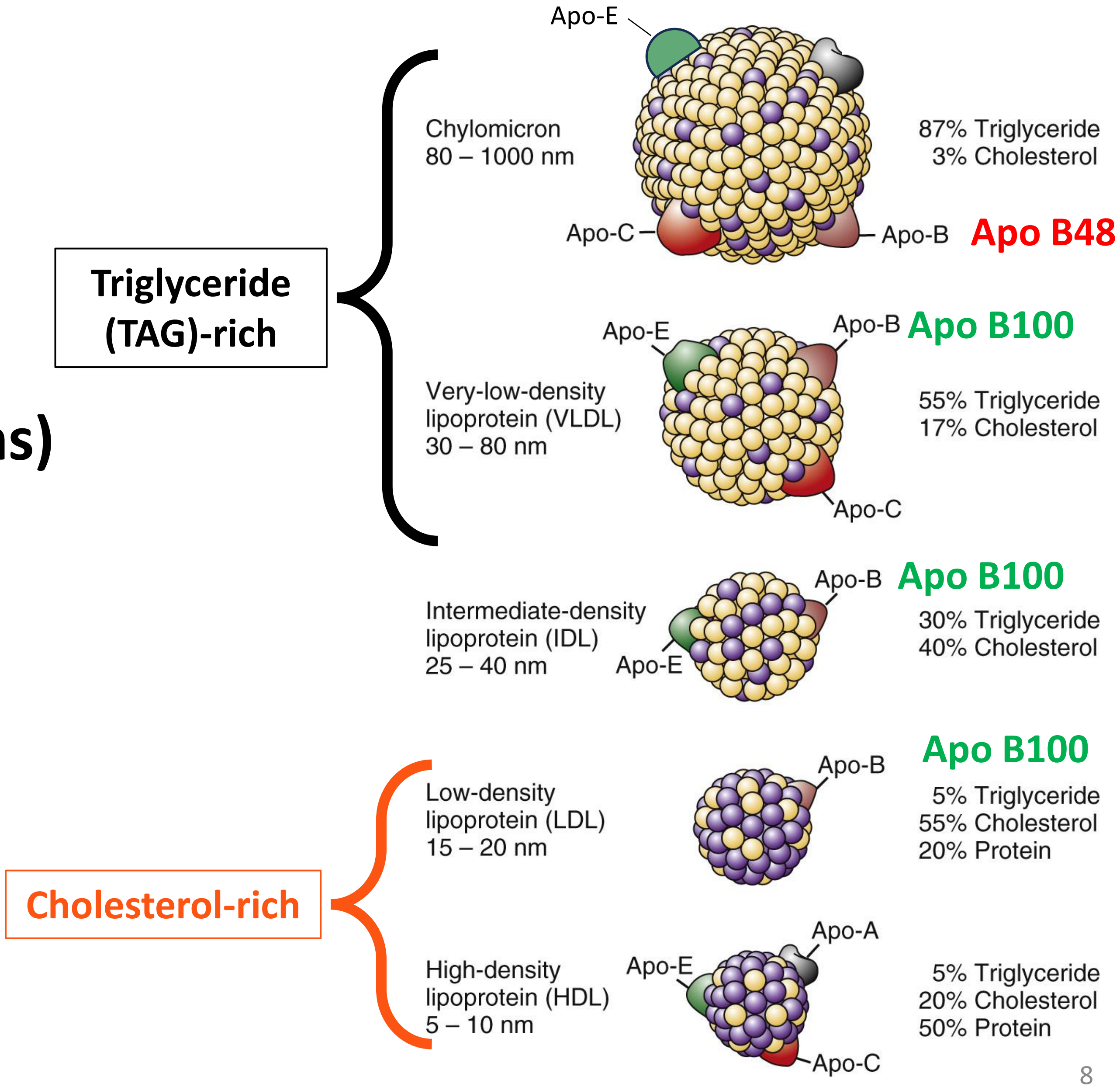
Cholesterol-rich





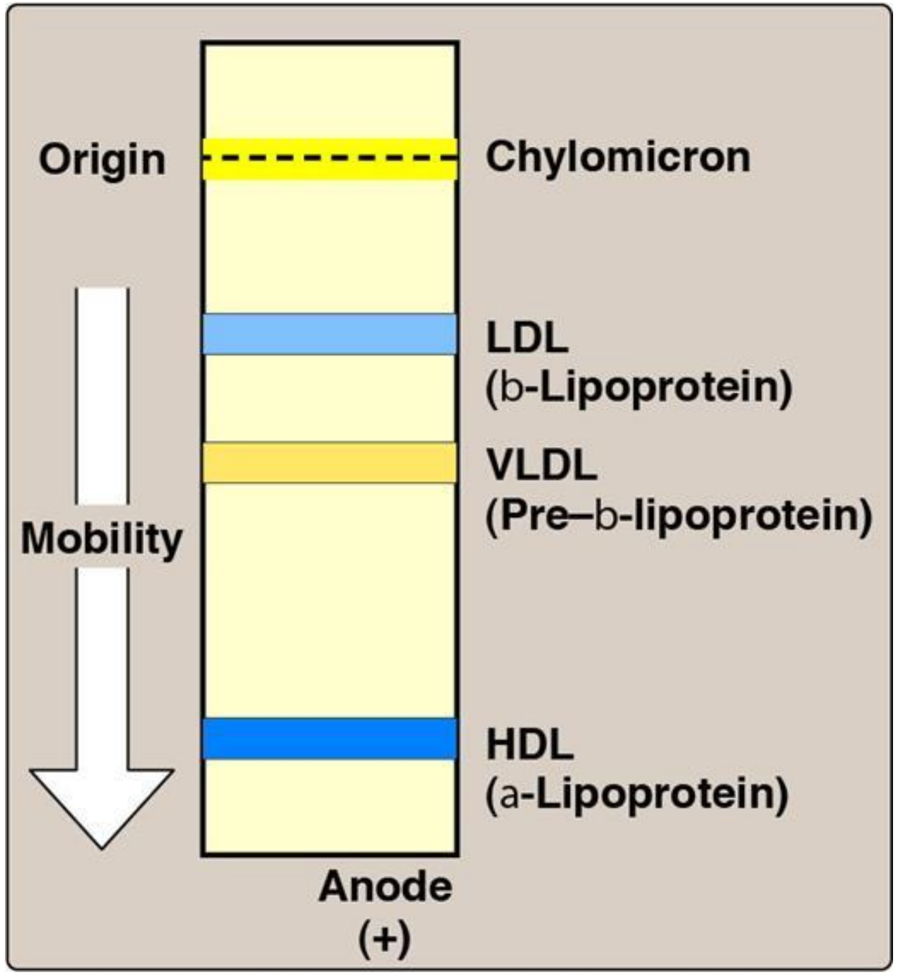
Classification of lipoproteins

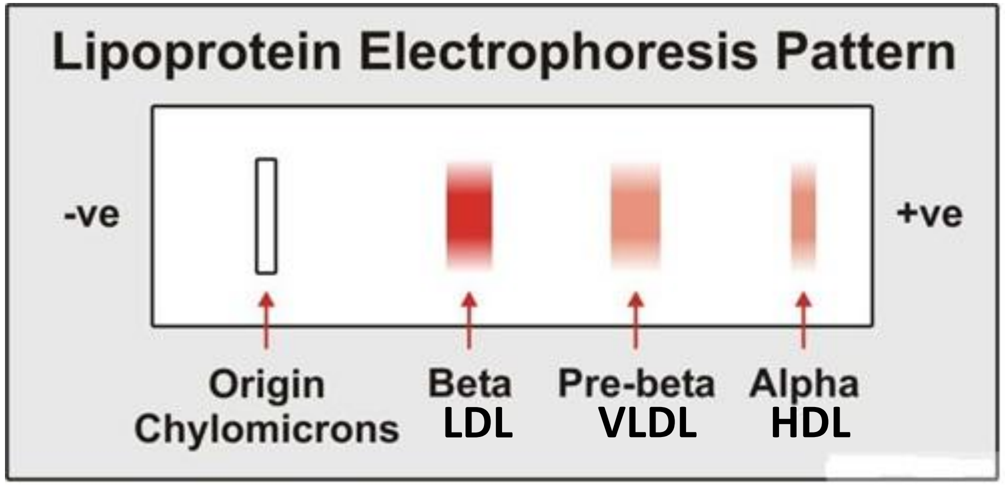
- **Lipoproteins**
  - **Composition and function**
  - **Apolipoproteins (Apoproteins) and function**
  - **Synthesis and fate (Metabolism)**
  - **Disorders related**



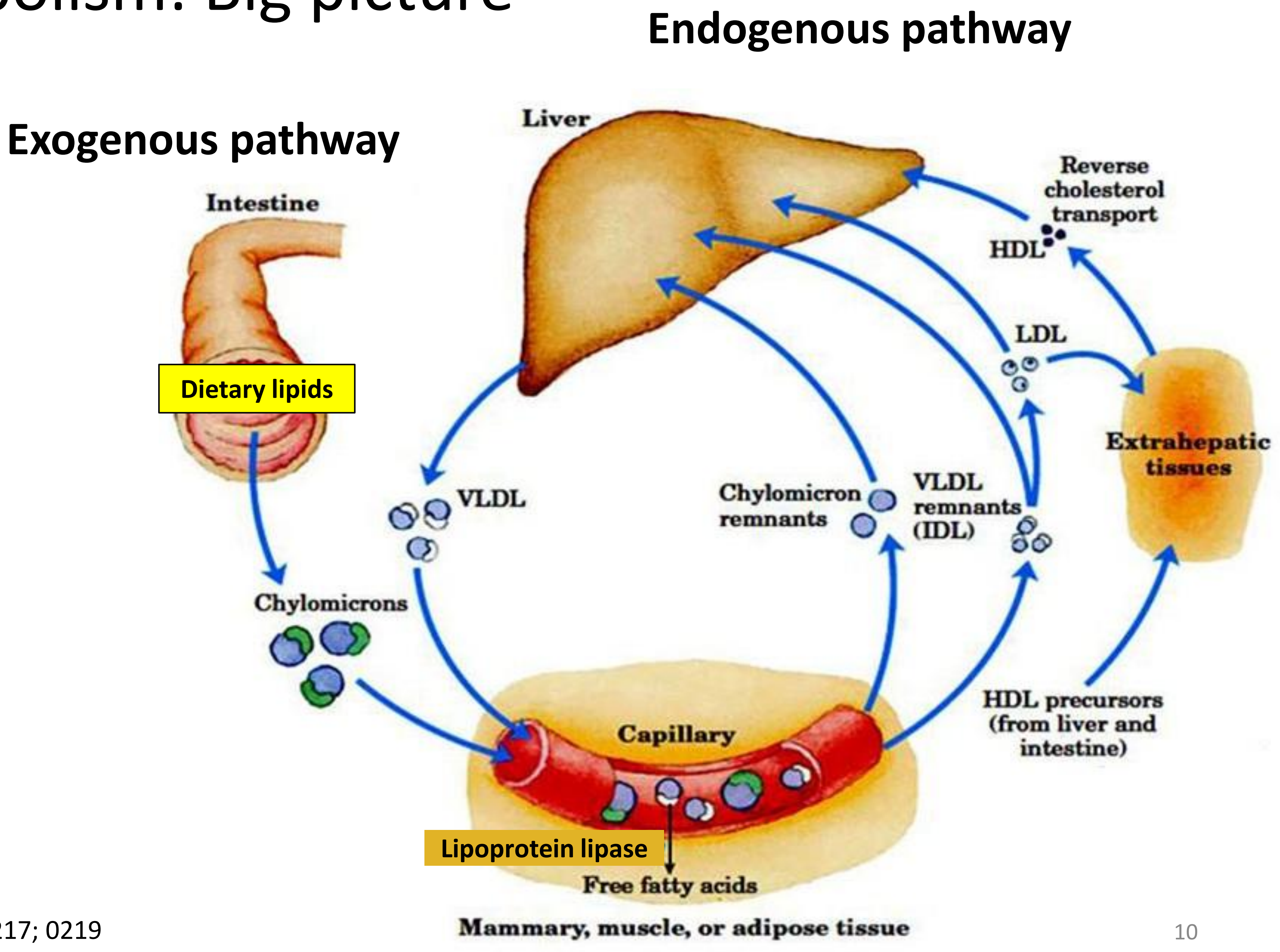
Electrophoresis of lipoproteins

- Based on the charge/mass ratio; HDL with highest protein content moves fastest
- Chylomicrons (origin)
- LDL (Beta/β-lipoprotein)
- VLDL (Pre-beta/β-lipoprotein)
- HDL (Alpha/α-lipoprotein)





Lipoprotein metabolism: Big picture



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Lipoprotein metabolism

- Exogenous pathway (transport of dietary lipids)
  - **Chylomicrons**: Transport **dietary TAG (triglycerides)** from intestine to adipocytes and muscle; TAG acted upon by **lipoprotein lipase**; Fatty acids formed, taken up and stored in adipose tissue or used for energy in muscle (Apo B48, C and E)
  - **Chylomicron remnants**: Transport dietary cholesterol to liver (Apo B48, E)
- Endogenous pathway (transport of endogenously synthesized lipids)
  - **VLDL**: Transport **TAG (triglycerides) from liver** to adipose tissue and muscle. TAG acted upon by **lipoprotein lipase**; Fatty acids taken up and stored in adipose tissue or used for energy in muscle (Apo B100, C, E)
  - **IDL**: (Apo B100, E)
    - About 50% converted to LDL; And, about 50% uptake by liver
  - **LDL**: Transports **cholesterol** to all tissues (Apo B100)
- **HDL:** Transports cholesterol from peripheral tissues to liver (reverse cholesterol transport) (Apo A, C, E)

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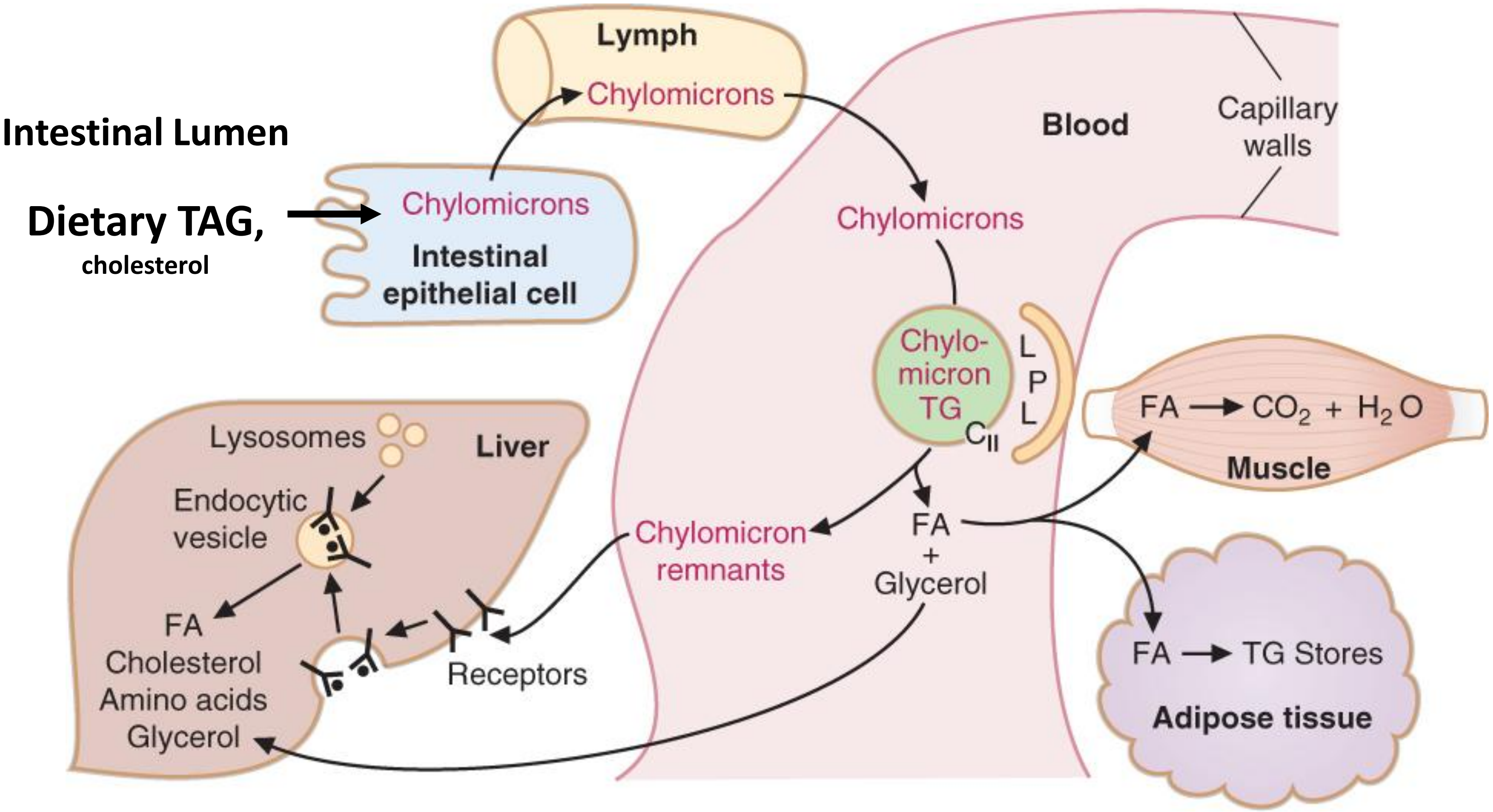
Chylomicron metabolism: Exogenous pathway

- **Chylomicrons**: Transport **dietary TAG (triglycerides)** from intestine to adipocytes (storage) and muscle (energy)
- **Chylomicron remnants**: Transport dietary cholesterol and fat-soluble vitamins to liver



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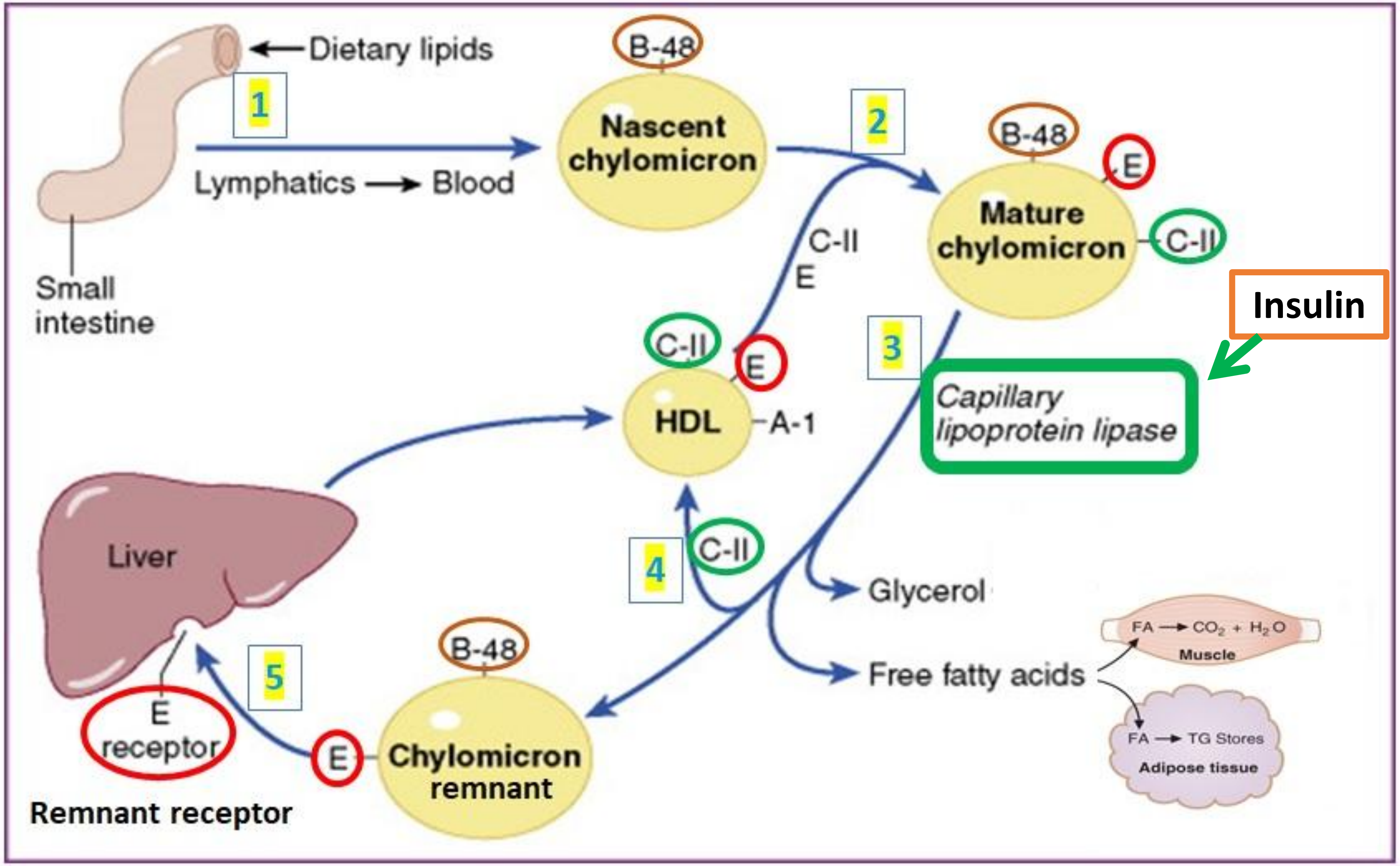
Chylomicron metabolism: Exogenous pathway



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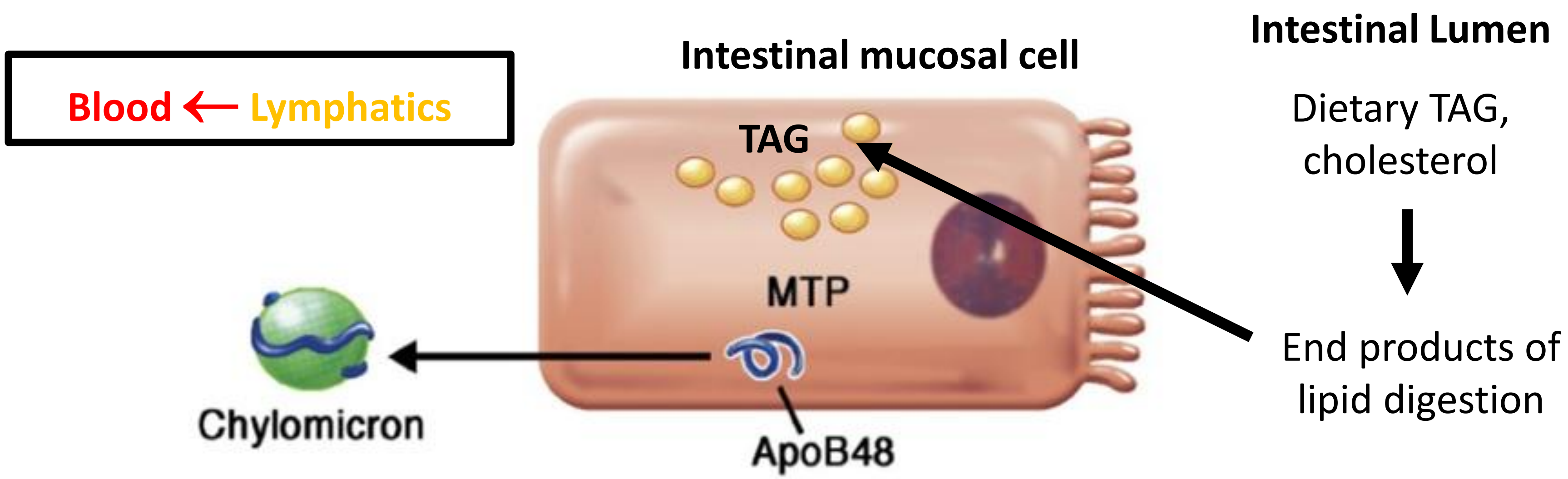
Chylomicron metabolism: Exogenous pathway

1. Nascent Chylomicrons **(ApoB48)** assembled in intestinal mucosal cell → Lymphatics of **intestine**→ Systemic circulation
2. **Apo C-II** and **Apo E** transferred from **HDL** to form **mature chylomicrons**
3. **Apo C-II** activates **Lipoprotein lipase (LPL)** in capillaries of adipose tissue and muscle; TAG → Fatty acids and glycerol; Fatty acids enter adipose tissue to be stored as TAG; Fatty acids used for energy in muscle
   - Chylomicron levels increase in circulation after a meal; Cleared by 4-6 hours after a meal due to activity of LPL
   - **Chylomicrons are not present in a normal person in the fasting state**
4. **Apo C-II** given back to HDL → Chylomicron remnants (Contain Apo B48 and Apo E)
5. Chylomicron remnants taken up by **remnant receptor** in liver which recognizes **Apo E**

Chylomicrons: Transport **dietary TAG** from intestine to adipocytes and muscle

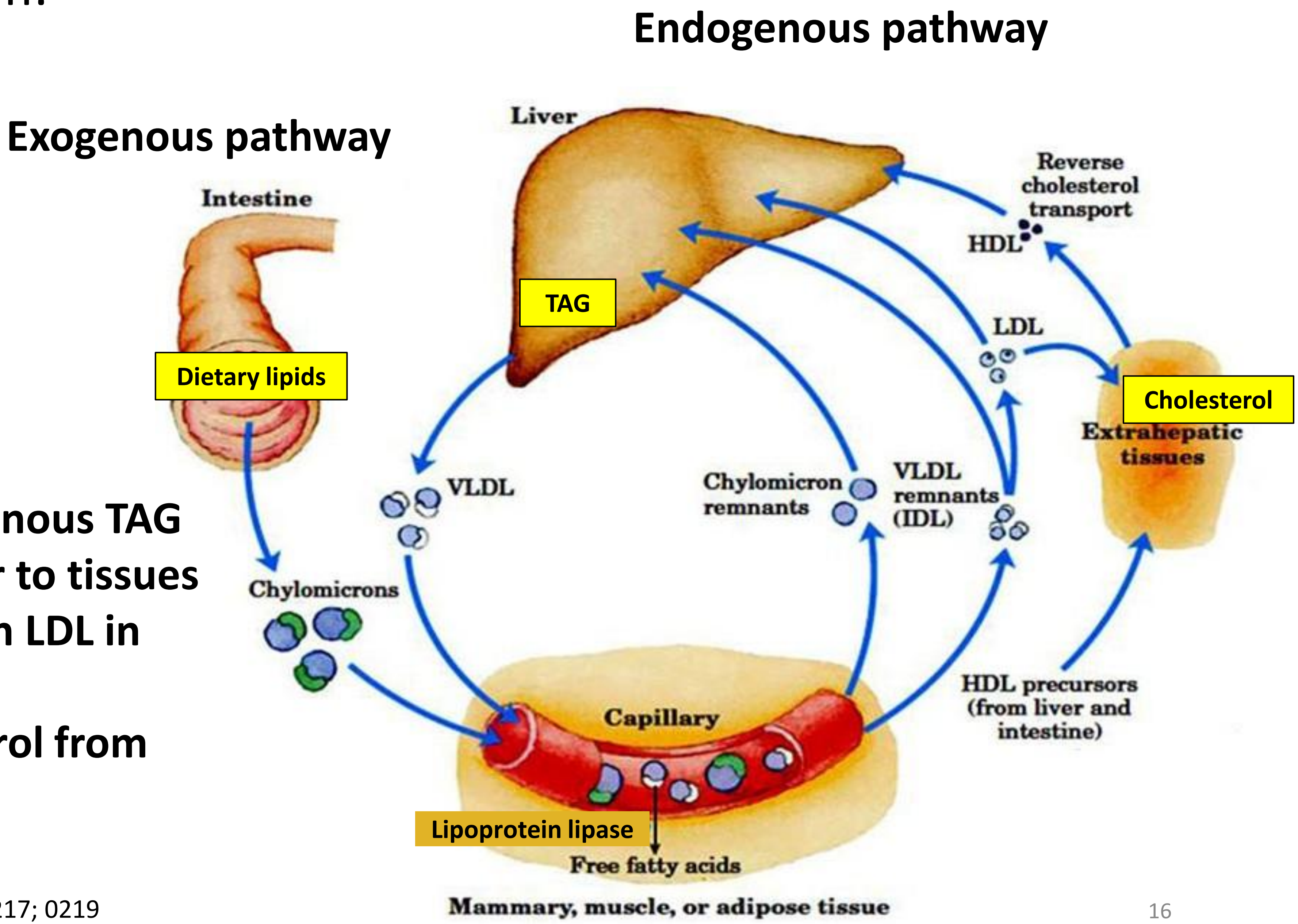
Chylomicron remnants: Transport **dietary cholesterol, fat-soluble vitamins to liver**

Microsomal triglyceride transfer protein (MTP)



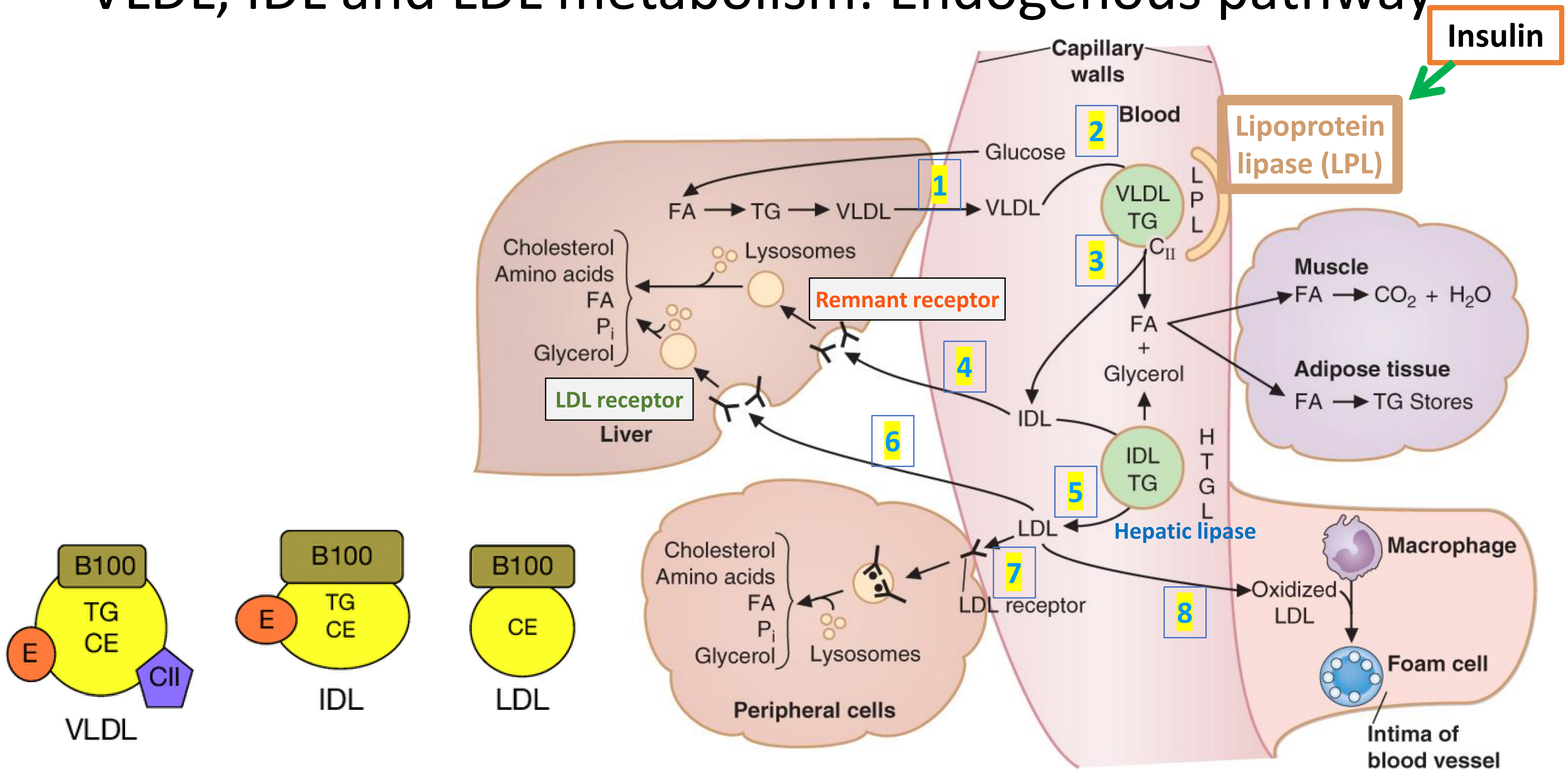
- Assists in addition of lipids (TAG/triglyceride) and in assembly and release of chylomicrons (intestinal mucosal cell) and VLDL (liver cell)
- **Abetaliproteinemia** (Inherited MTP deficiency) leads to
  - Low chylomicron levels (Poor dietary lipid absorption) → Fat in stools (steatorrhea); Fat soluble vitamin deficiency
  - Low VLDL levels → Low serum triglyceride levels
  - Low LDL levels → Low serum cholesterol levels

Lipoprotein metabolism: Endogenous pathway

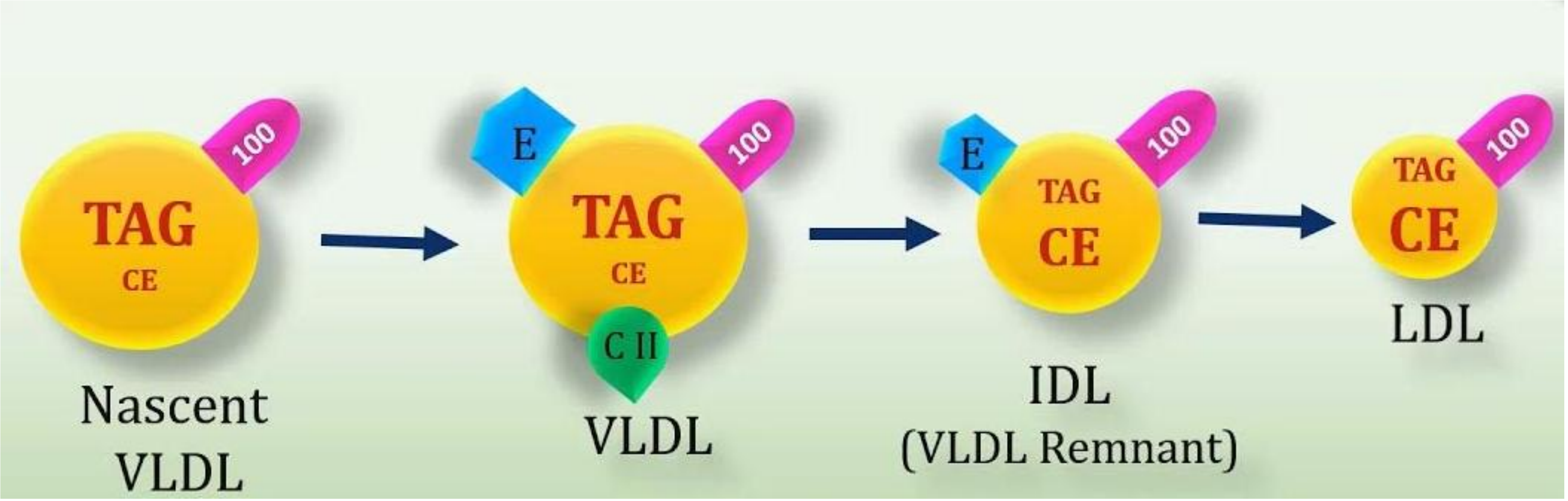


- **VLDL transports endogenous TAG (triglycerides) from liver to tissues**
- **VLDL forms IDL and then LDL in circulation**
- **LDL transports cholesterol from liver to tissues**

VLDL, IDL and LDL metabolism: Endogenous pathway



VLDL, IDL and LDL metabolism: Endogenous pathway



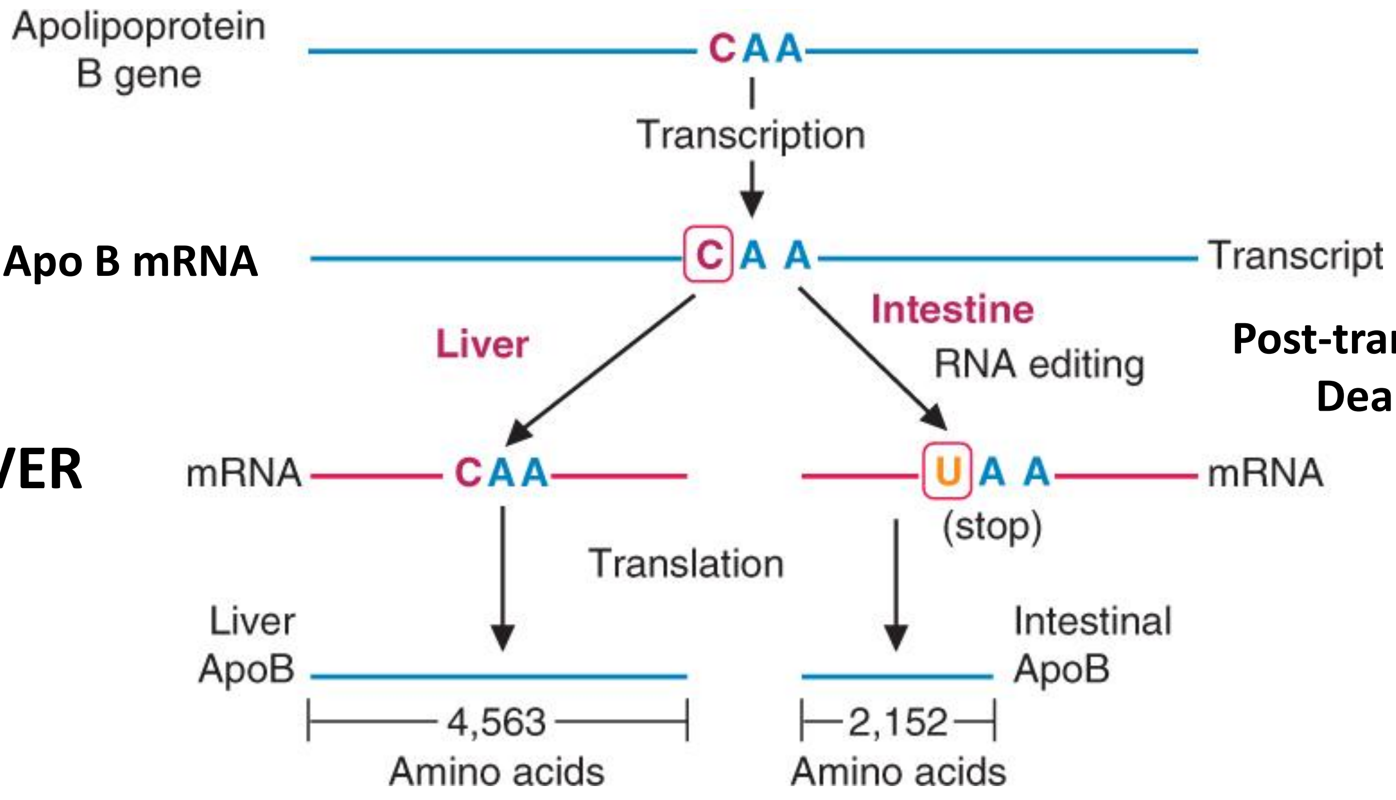
VLDL, IDL and LDL metabolism: Endogenous pathway

1. Release of nascent VLDL (ApoB-100) from liver (contains endogenous TAG)
2. Formation of **mature VLDL**: **Apo-C-II** and **ApoE** given by HDL
3. Apo C-II activates **Lipoprotein lipase (LPL)**: TAG is removed and VLDL is converted to **IDL** (remnant VLDL); Apo C-II given to HDL
4. 50% of **IDL** is taken up by liver by the **remnant receptor** that recognizes **ApoE**
5. 50% of IDL converted to **LDL**; ApoE given to HDL and TAG removed by **hepatic lipase**
6. About 50-60% of **LDL** taken up by liver via **LDL-receptor** which recognizes **Apo B100**
7. About 50% of LDL taken up by extrahepatic tissues via LDL-receptor which recognizes **Apo B100**
8. **Oxidized LDL** taken up by macrophages in blood vessels via **SR-A** receptors (Scavenger receptor)

VLDL transports endogenous TAG from liver to tissues

LDL transports cholesterol from liver to tissues

Apo B48 and Apo B100: RNA editing



Apo B mRNA

Post-transcriptional RNA editing: Deamination of Cytidine

LIVER

INTESTINE

Apo B100

Apo B48: Shorter protein in INTESTINE

VLDL, IDL and LDL

Chylomicrons and Remnant chylomicrons

Hypobetalipoproteinemia: Mutation in Apo B gene

- Low chylomicron levels (Poor dietary lipid absorption) → Fat in stools (steatorrhea); Fat-soluble vitamin deficiency
- Low VLDL levels → Low serum triglyceride levels
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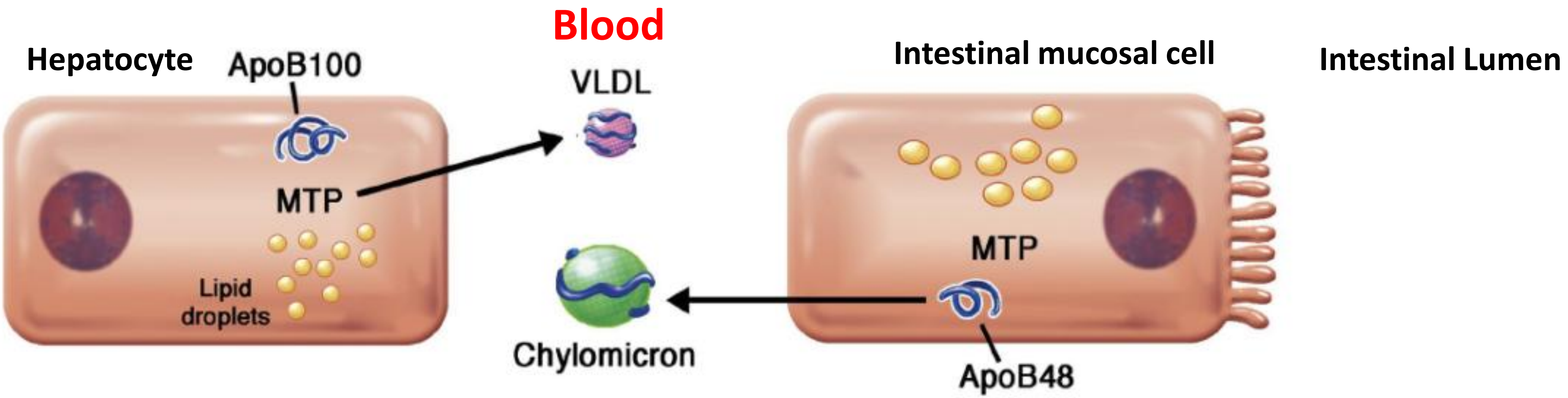
Microsomal triglyceride transfer protein (MTP)



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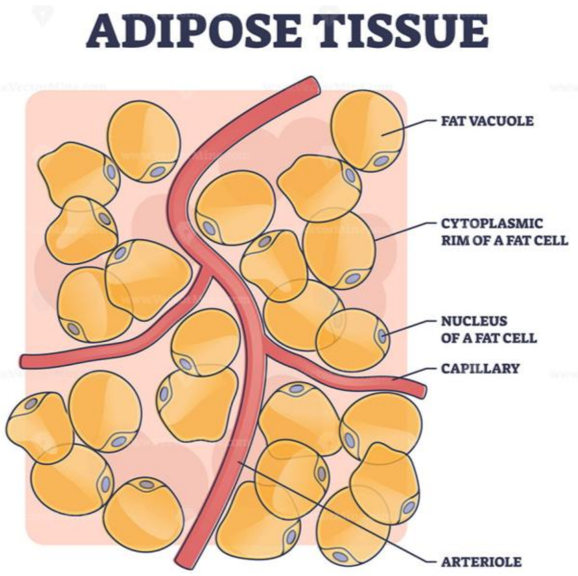
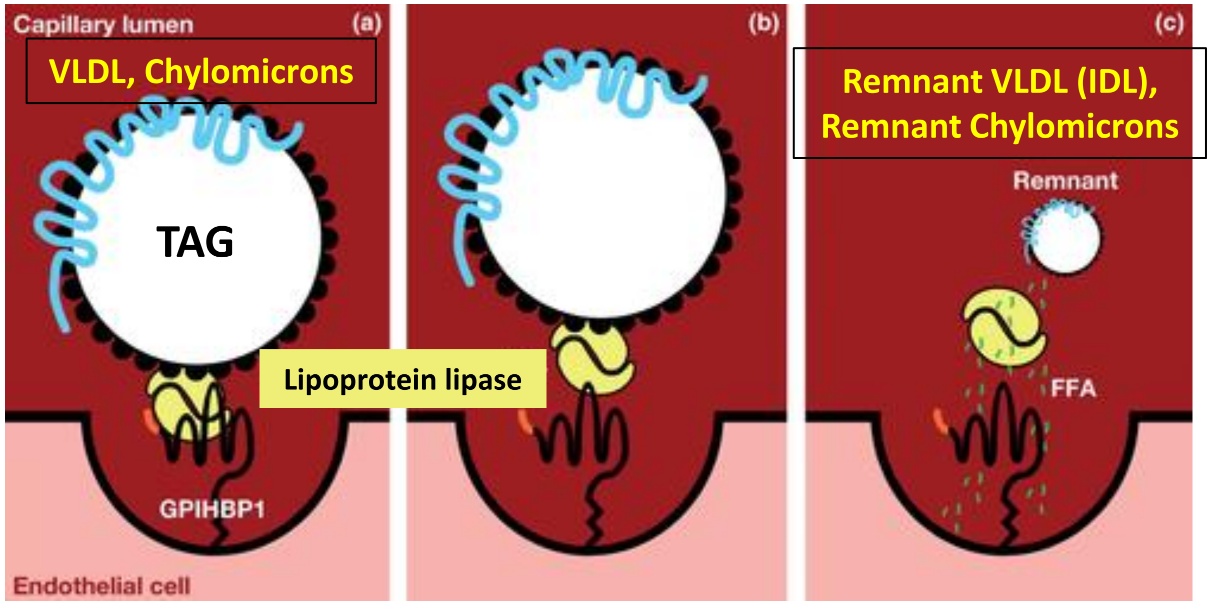
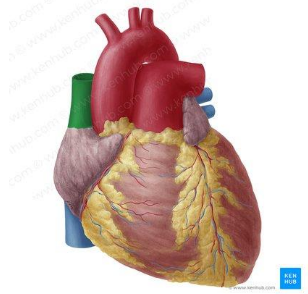
Lipoprotein lipase

VLDL, Chylomicrons

Triglycerides (TAG) → **Lipoprotein lipase** (Apo C-II) → **Fatty acids** → **Stored as TAG (adipose)** / **Energy**

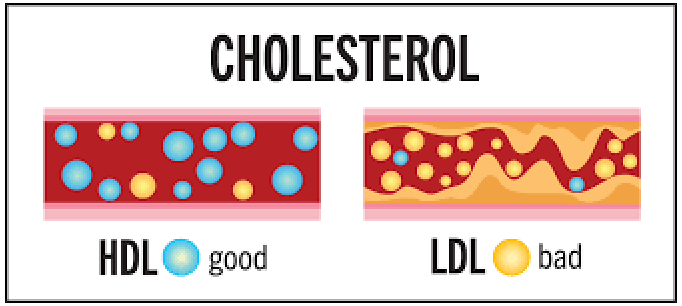




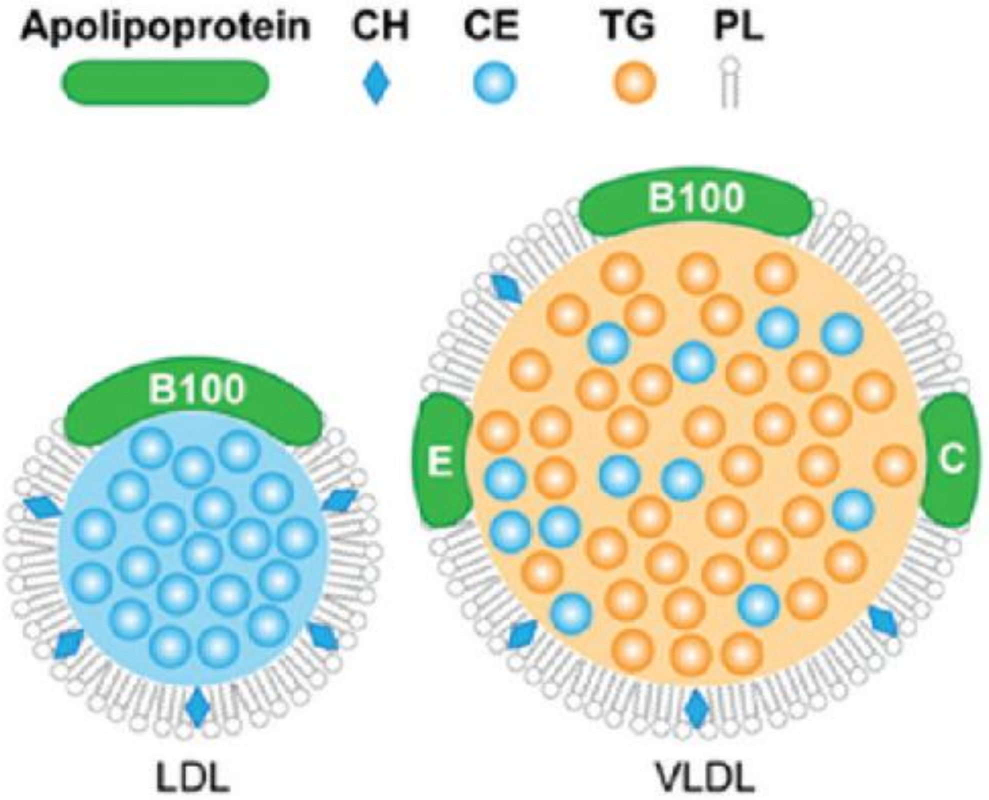


- Produced by adipocytes and muscle (skeletal and cardiac); Anchored to **endothelial cells** in capillaries
- Activated by **Apo C-II** (in Chylomicrons and VLDL)
- **Stimulated by insulin**; Promotes **storage of TAG (triglycerides)** in adipose tissue
- **Insulin resistance** (Obesity or type-2 diabetes mellitus): Low LPL activity → Elevated serum triglycerides (↑VLDL)
- **Inherited LPL deficiency or ApoC-II deficiency**: Increases Serum chylomicrons or VLDL or both; Elevated serum triglycerides

Low density lipoprotein (LDL)



- Formed in circulation from VLDL → IDL → LDL
- TAG in VLDL and IDL removed by **Lipoprotein Lipase (LPL)** and Hepatic lipase
- Contains only **ApoB100**
  - ApoC and ApoE given to HDL
- **LDL transports cholesterol to tissues**
- Cholesterol used for synthesis of membranes, steroid hormones, bile acids (liver)
- 50-60% taken up by liver; 50% taken up by extra-hepatic tissues
- Uptake by **receptor mediated endocytosis**
- **LDL receptor recognizes ApoB100**
- **LDL receptor mutations: Familial hypercholesterolemia**



Cholesterol rich **Triglyceride (TAG) rich**

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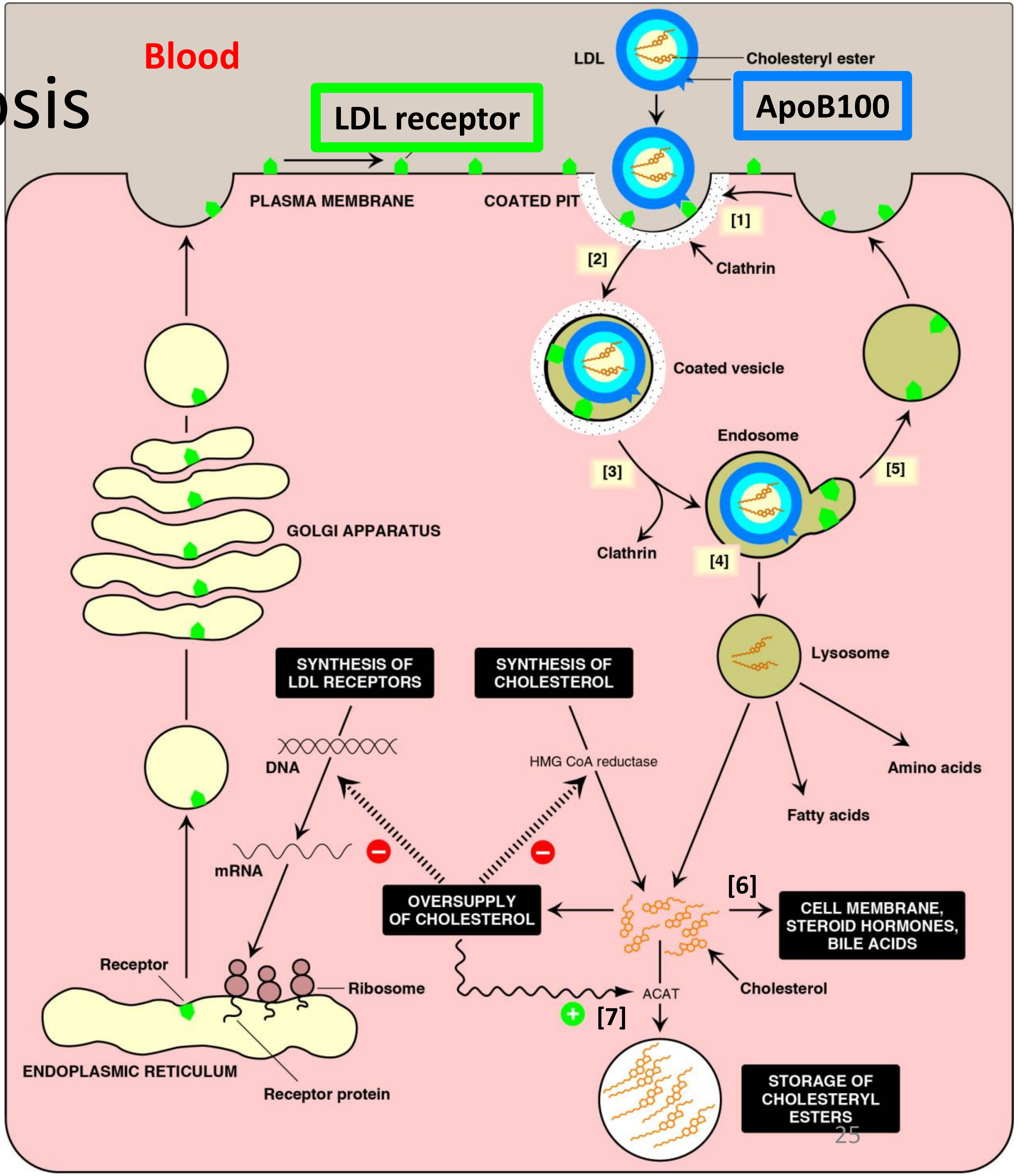
LDL receptor mediated endocytosis

1. **LDL receptors** present in plasma membrane of all cells (Clathrin coated pits)
   - **LDL receptor mutations → Familial hypercholesterolemia**
2. LDL receptor binds to ApoB100 on LDL; **Receptor mediated endocytosis**
3. Endosome fuses with lysosome
4. Forms free cholesterol,
5. LDL receptor recycled to cell membrane

Fates of intracellular cholesterol:

6. Cell membrane, steroid hormone synthesis, bile acid synthesis (liver)
7. Forms cholesterol esters by ACAT (Acyl CoA cholesterol acyl transferase)



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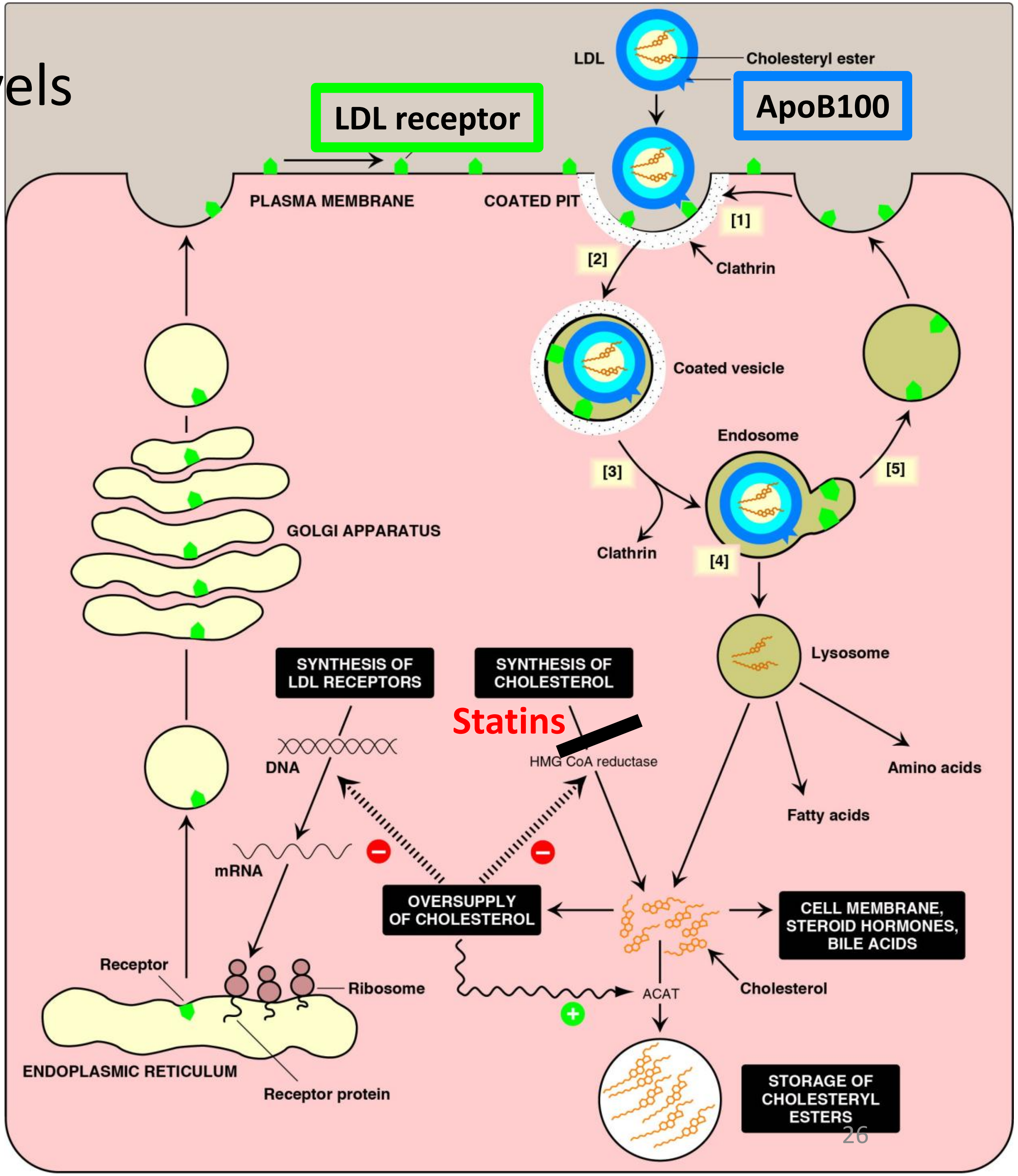
Regulation of intracellular cholesterol levels

- Increased intracellular free cholesterol
  - Activates ACAT → Forms cholesterol esters (Cholesterol + fatty acid)
  - Inhibits cholesterol synthesis; Inhibits HMG CoA reductase
  - Inhibits synthesis of LDL receptor
- **Statins**
  - Competitive inhibitors of **HMG CoA reductase** (Inhibit cholesterol synthesis)
  - Stimulate LDL receptor synthesis → Facilitate increased uptake of LDL from blood → Decreases serum LDL cholesterol and total serum cholesterol levels



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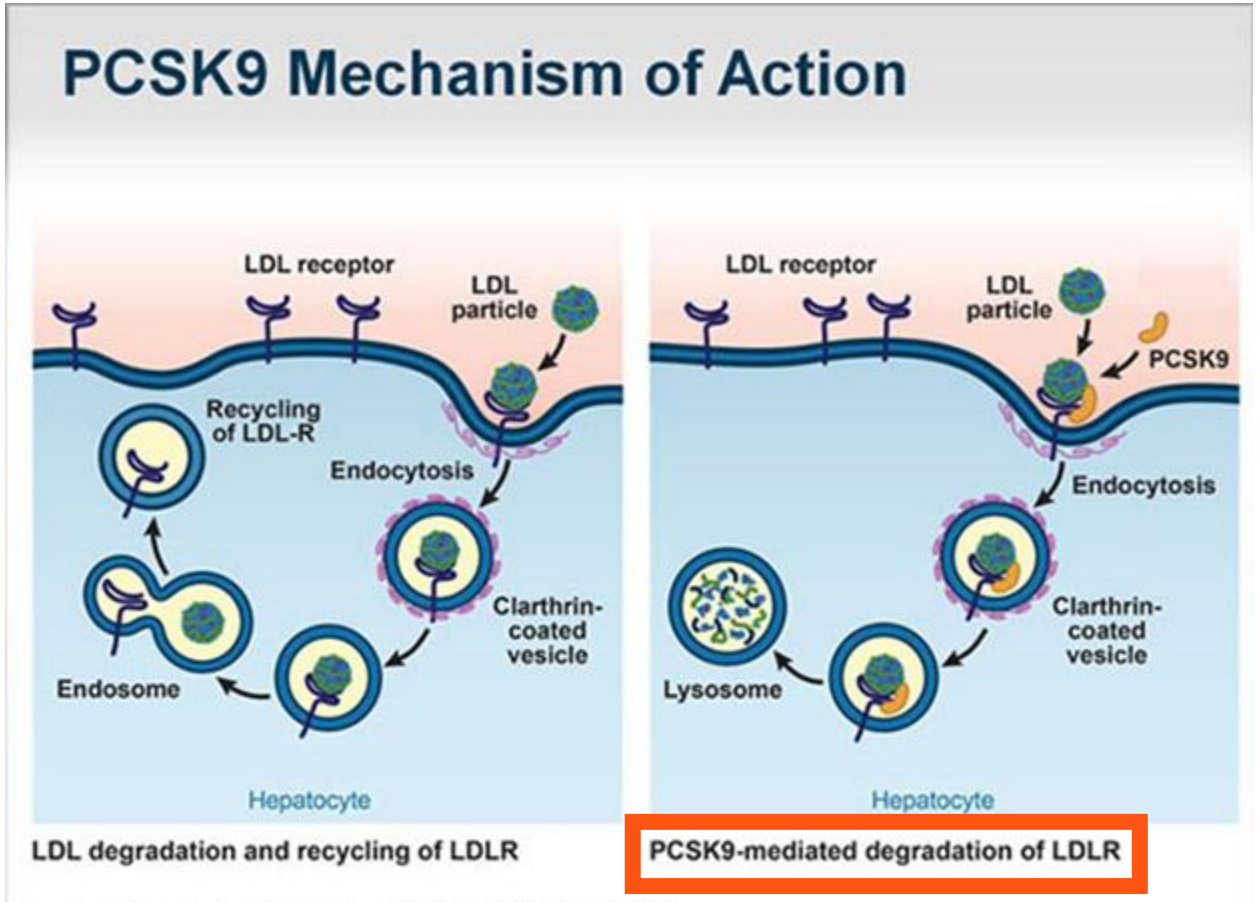
PCSK9 (Proprotein convertin subtilisin kexin type 9 serine protease)

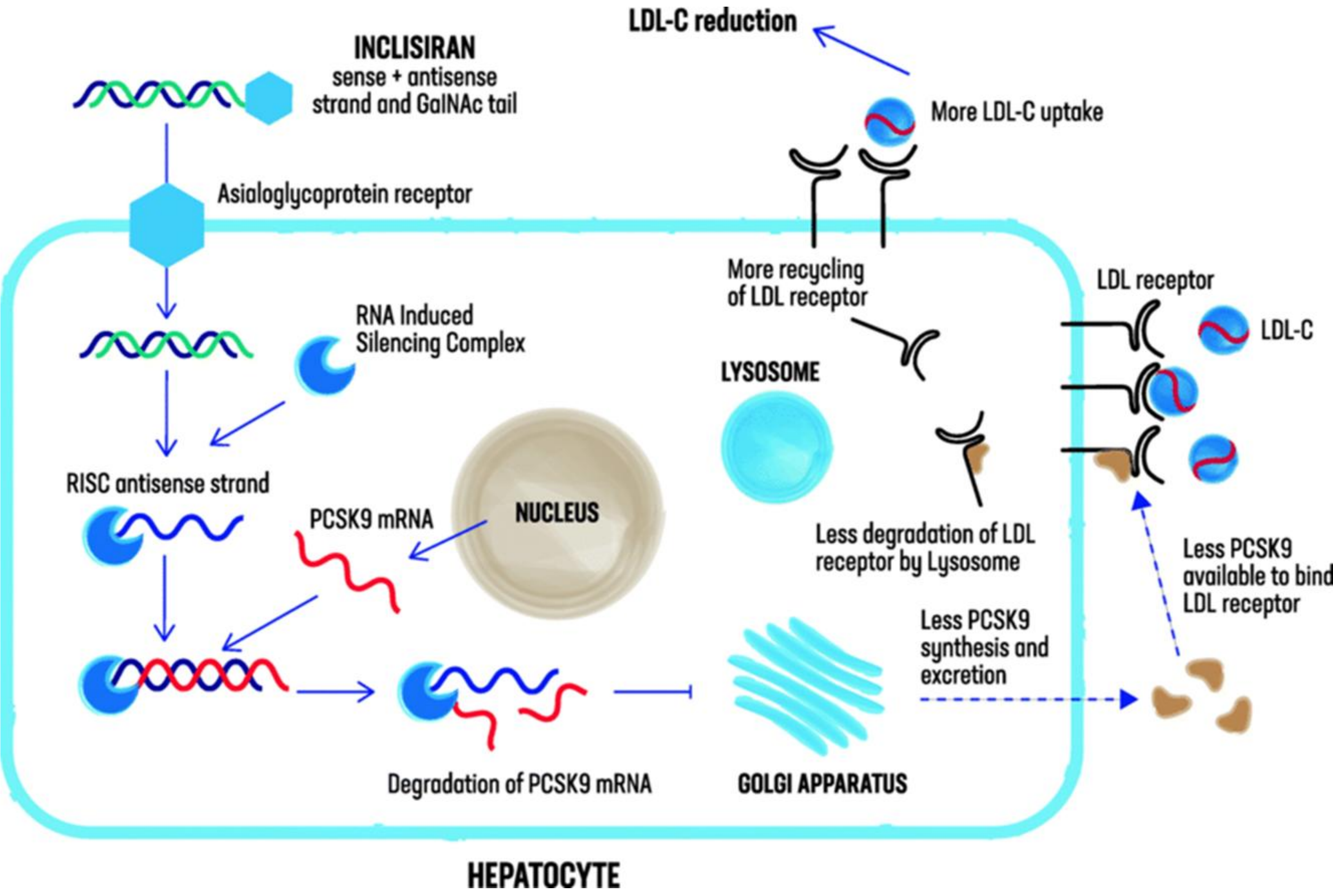


Lambert G, et al. *J. Lipid Res.* 2012;53:2515-2524.[6]

- When PCSK9 binds to LDL receptor during endocytosis → **Degradation of LDL receptor (LDL receptor not recycled to cell membrane)**
- PCSK9 synthesized and released by hepatocytes
- **High levels of PCSK9 →Less recycling of LDL receptors → Increases LDL cholesterol levels in blood**
- PCSK9 inhibitors: used in treatment of hypercholesterolemia **(promote LDL receptor recycling)**

Inclisiran: LDL-Lowering siRNA Therapeutic Agent

- Inclisiran: novel siRNA drug **reduces** proprotein convertase subtilisin/kexin type 9 **(PCSK9) gene expression**
- Reduces LDL-C levels



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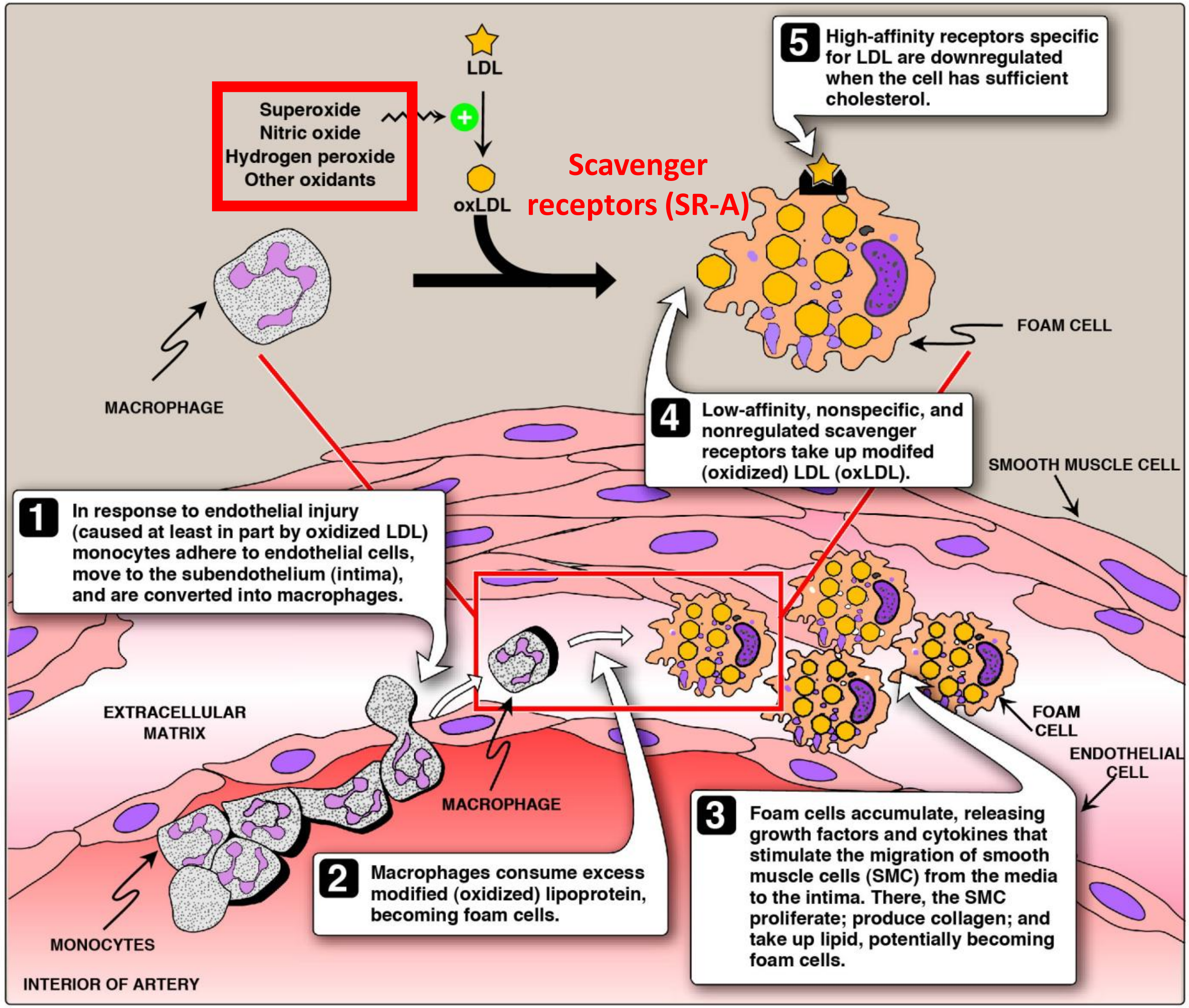
[Inclisiran: a new generation of lipid-lowering siRNA therapeutic - PMC (nih.gov)]

Oxidized LDL and atherosclerosis

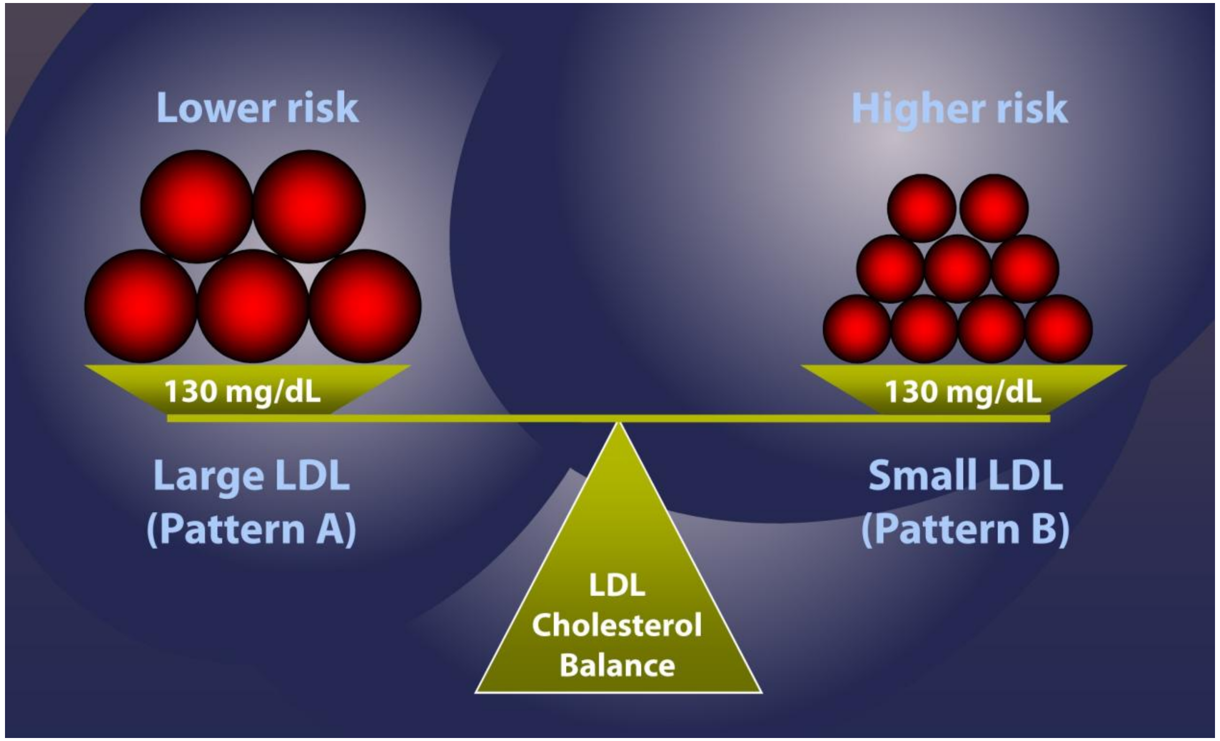
- LDL in circulation and in the sub-endothelial space (intima) becomes oxidized by **free radicals/ oxidants**
- Oxidized LDL taken by macrophages via **scavenger receptors (SR-A)**
- Macrophage→ Foam cell → Release cytokines → Migration of smooth muscle cells → → Atherosclerosis
- **Higher LDL cholesterol levels→ Higher oxidized LDL and higher risk of atherosclerosis**



LDL-B (Small LDL)

- Highly enriched in cholesterol
- Higher risk of entering subendothelial space (intima)
- **Higher oxidation and higher risk of atherosclerosis**
- Diet: Trans-fats and saturated fats increase LDL-B

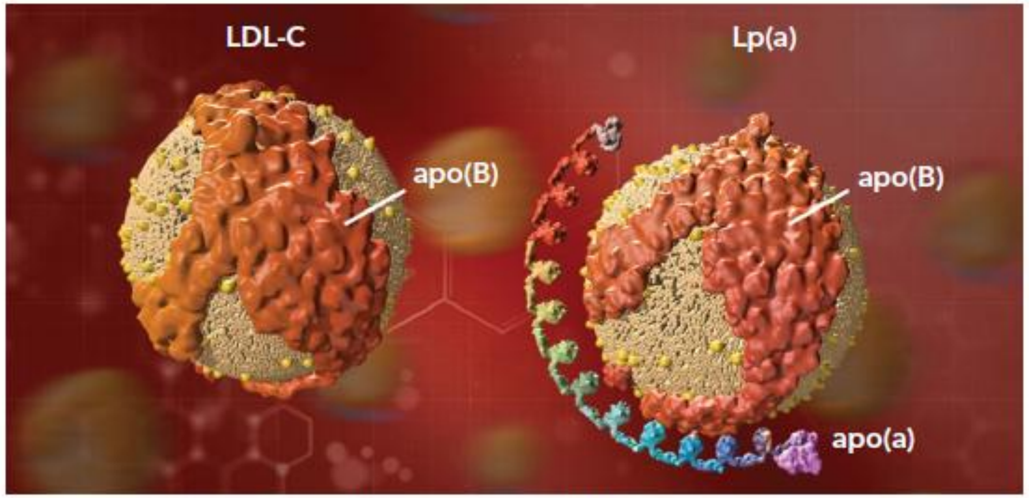
Atherosclerotic cardiovascular disease (ASCVD) risk



Lipoprotein (a) and atherosclerosis

- Lipoprotein (a) is an LDL particle with a covalently attached **Apo(a)**
- Levels are Genetically determined
- **Correlates with higher risk of atherosclerotic cardiovascular disease (ASCVD)**
  - Higher risk of oxidation and foam cell formation
- Plasminogen-like domains which **inhibit fibrinolysis** (facilitate thrombus formation)

Figure 1: Low-density Lipoprotein Cholesterol and Lipoprotein(a)



apo(a) = apolipoprotein(a); apo(B) = apolipoprotein B; LDL-C = low-density lipoprotein-C; Lp(a) = lipoprotein a. Source: Anfinsen et al. 1994.[88] Adapted with permission from Elsevier.